

Session 7: What is Generative AI?

So far, AI has been **recognizing** things - cats, dogs, hand gestures. Today, we explore AI that **CREATES** things. Stories. Pictures. Music. Videos. AI can make all of it.

Welcome to Generative AI - the most magical-feeling part of this course. ✨



Objectives:

By the end of this session, you'll be able to:

01

Define It

Say what **Generative AI** is in your own words

02

Compare

Tell the difference between AI that **recognizes** vs AI that **creates**

03

Explain

Describe in simple words HOW Generative AI makes new stuff

04

Spot the Types

Identify the main types: **text, images, audio**

05

Use a Prompt

Give AI an instruction and make it create something for you

Recap & Homework Share

Welcome back! Let's see your dataset homework.

Homework Questions

- What dataset did you find?
- What could an AI learn from it?
- What might go wrong with it?

Quick Check from Last Session

- What's overfitting?
- Why is more data usually better?
- What's a dataset?



Imagine This... 🎂

Imagine it's your best friend's birthday next week. You want to do something really special. But what?



A Personalized Song

Make them a song about your inside jokes



A Custom Birthday Card

Draw a card featuring their favorite animal



A Funny Poem

Write them a hilarious birthday poem

👉 Spoiler: You can. All of it. Using AI. That's Generative AI — and today, you'll use it.

What is Generative AI?

AI that CREATES new stuff.

Not just recognizes things. Not just sorts things. **Makes brand new stuff.**



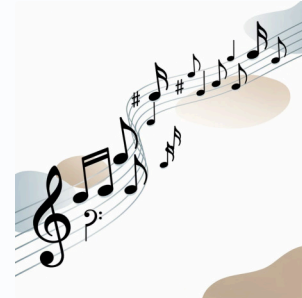
Stories & Poems

Full narratives, rhymes, jokes, and messages written from scratch



Pictures & Drawings

Original artwork, illustrations, and designs generated on demand



Songs & Sound Effects

Full tracks, melodies, and audio created from a simple description



Videos & Animations

Moving images and animated scenes generated by AI

- ① You used one in Session 1 - Google Storybook (the AI that made stories with you, remember?). That was Generative AI making stories.

"Regular" AI vs. Generative

Two big jobs AI can do — and they're very different:

Regular AI (Recognizing)

Like a **detective**

- Answers: "What is this?"
- Sorts photos into folders
- Checks if a comment is mean
- Tells if an email is spam

Generative AI (Creating)

Like an **artist**

- Answers: "What can I make?"
- Draws a brand new photo
- Writes a kind comment
- Writes an email for you

You've spent 6 sessions learning about recognizing AI. This session is creating AI.



Mini Activity 1 – Recognize or Create?

For each task below, decide: is it **RECOGNIZING** or **CREATING**?

Task	Your Answer
Writing a birthday poem for your friend	?
Sorting all photos of your dog into one folder	?
Drawing a unicorn wearing a party hat	?
Spam filter in email	?
Making a birthday song about your inside jokes	?
Counting how many people are in a class photo	?
Writing a birthday card message	?
Telling if a comment online is friendly or mean	?

Mini Activity 1 – Answers

Task	Answer
Birthday poem	 Creating
Sorting photos	 Recognizing
Drawing a unicorn	 Creating
Spam filter	 Recognizing
Birthday song	 Creating
Counting people	 Recognizing
Birthday card message	 Creating
Friendly or mean?	 Recognizing

 Creating (Generative)

If the AI's job is to **MAKE something new**

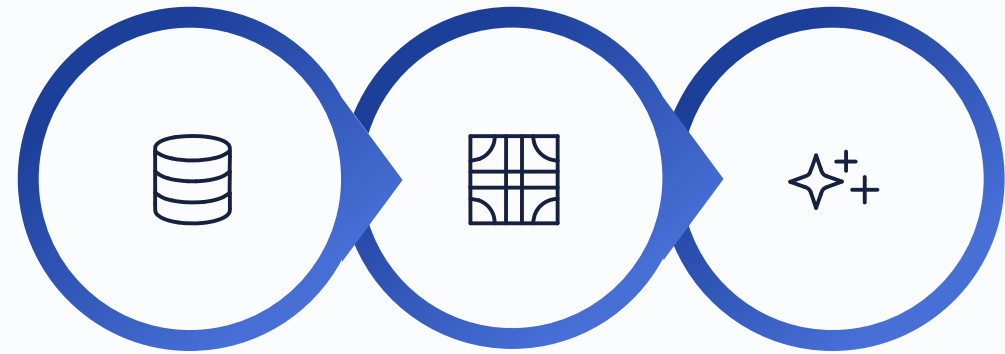
 Recognizing

If the AI's job is to **LABEL or SORT** existing stuff



How Does Generative AI Work?

It's simpler than you think — just 3 steps:



Learn

Find Patterns

Remix

That's it. **Learning + Patterns + Remixing.** Every piece of AI-generated content — a poem, a painting, a song — comes from this same three-step process happening billions of times over.

The Remix Artist Idea 🎧

Here's the best way to think about Generative AI:



They Listen

A remix artist listens to thousands of songs — pop, rock, jazz, hip-hop



They Learn Patterns

How melodies work, what chords sound cool, how rhythms feel



They Create Something New

A brand new song — all their own — but built from everything they listened to

- ① That's exactly what Generative AI does. It's a remix artist. It learned from millions of examples — and now it can make NEW stuff that's a "remix" of all of it.



The Most Important Thing About Generative AI

Earlier we said Generative AI is like an artist. But there's a **HUGE** difference: A real artist understands what they're making. Generative AI does NOT.

When AI writes you a birthday poem:

Doesn't Know Birthdays

Doesn't Feel Friendship

Doesn't Know Why It's Special

It just knows that in millions of texts, the words "birthday," "happy," "friend," and "cake" usually go together. **It's a pattern-remixer. Not a thinker.**

What you see

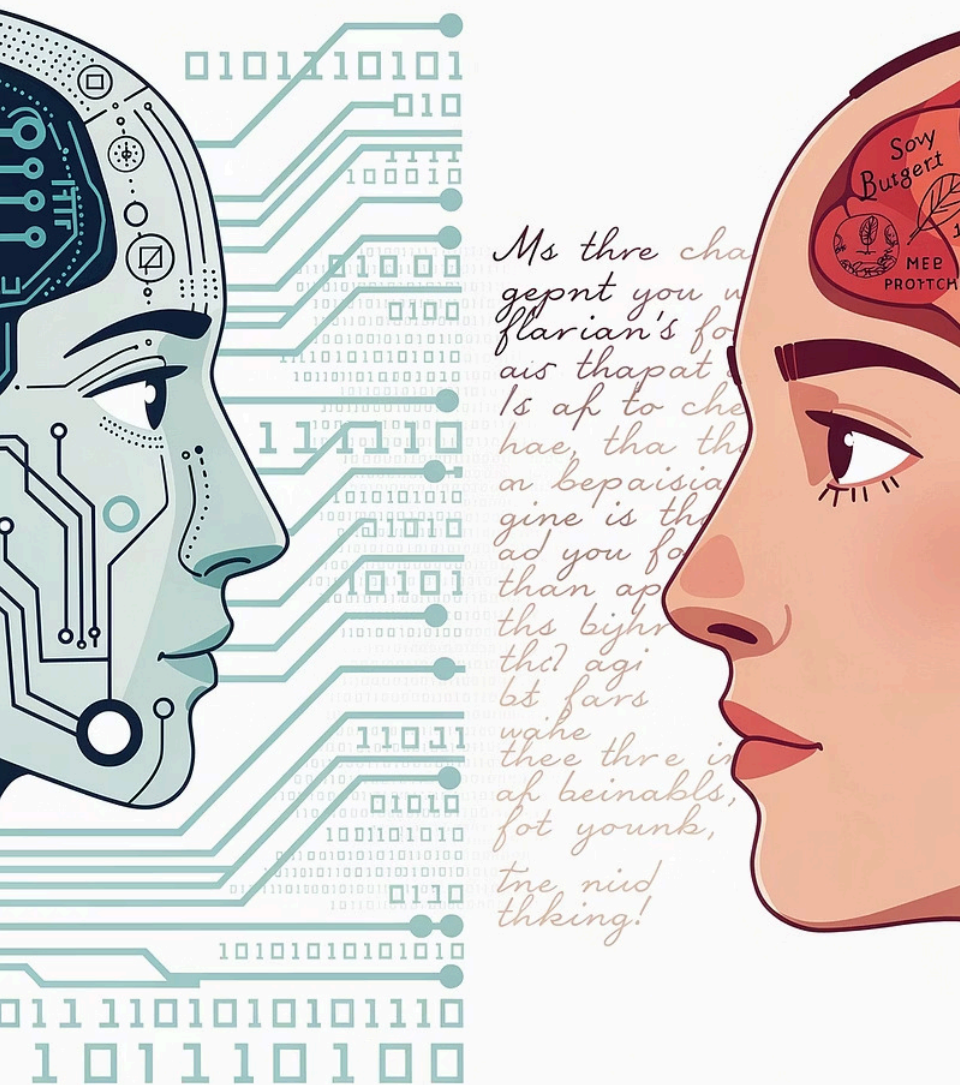
A warm, personal
birthday poem



What actually
happened

Pattern-matching
across millions of
texts

⚠ This is why AI can sometimes confidently say things that are totally wrong — it's just remixing patterns, not checking if they're true. We'll dive into this in a session called "AI Hallucinations" coming up!



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Three Types of Generative AI

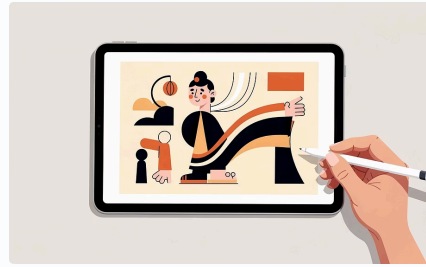
Generative AI comes in 3 main flavors — each trained on a different kind of data:



Text

Stories, poems, messages, jokes

- ChatGPT
- Gemini
- Google Storybook



Images

Art, cards, posters, drawings

- Gemini
- Canva AI
- DALL-E



Audio

Songs, sound effects, voices, and music.

- Suno (songs!)
- ElevenLabs (voices!)

Each one learned from millions of examples in that type. The more examples it saw, the better it got.



Generative AI Trained on... A LOT

To get good, Generative AI needs **HUGE** amounts of training data:

- Text AI

Trained on **BILLIONS** of pages — books, articles, websites. That's more text than any human could read in 1,000 lifetimes!

- Image AI

Trained on **MILLIONS** of pictures — photos, paintings, drawings, and designs from across the internet

- Audio AI

Trained on **MILLIONS** of hours of sound — songs, voices, podcasts, and sound effects



Remember: **Garbage in = Garbage out!** If the training data was bad → the AI will create bad stuff.

Example: If an image AI is trained only on cartoon drawings, it'll struggle to make a realistic photo.

Generative AI in Your Everyday Life

You probably use Generative AI **without realizing it!**



Snapchat AI

Chats with you (Generative AI!)



Snapchat Lenses

AI-made backgrounds and effects



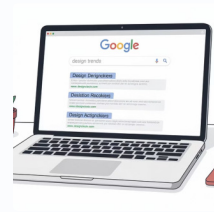
Instagram Stories

"Imagine a background of..." — AI generates it



Spotify AI DJ

AI-narrated music sessions personalized for you



Google Search

AI summaries at the top of your results



TikTok Filters

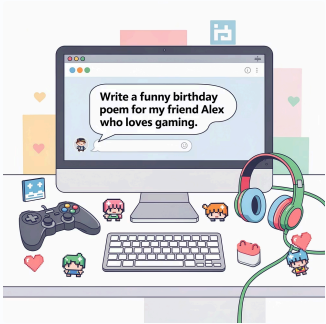
AI-generated visual effects on your videos

✓ Generative AI isn't future tech. It's already in your phone.

The Magic Word — PROMPT ✨

Prompt = your instruction to AI.

It's literally the words you type to tell AI what to make. The better your prompt → the better the result.



Text Prompt Example

"Write a funny birthday poem for my friend Alex who loves gaming."



Image Prompt Example

"Make a picture of a unicorn wearing a party hat."



Audio Prompt Example

"Write a 30-second birthday song about pizza."

📄 **Big idea:** YOU are in charge. The prompt shapes everything. AI is your assistant — you are the director.

Mini Activity 2 – Plan a Birthday Surprise! 🎉

Time to plan something cool. Pick **ONE** for your best friend's birthday:

TEXT Idea

- A funny birthday poem
- A heartfelt birthday message
- A short story where your friend is the hero

IMAGE Idea

- A custom birthday card with their favorite animal
- A poster about your friendship
- A digital gift tag

Your Quick Plan

01

What Type?

Choose **Text** or **Image** — which one excites you most for your friend's birthday?

02

What's the Idea?

Pick your specific idea from the list — a poem, a card, a story, a poster, or a gift tag. Make it personal!

03

Write a First Prompt

Draft your very first prompt to try. Describe what you want to create in 1–2 sentences. Be specific and creative!



How to Write a Good Prompt – Quick Tips

3 tips to make your prompt better every time:

01

Be Specific

~~"A birthday card"~~

"A birthday card with a dragon wearing a party hat, eating cake"

"Write a birthday poem."

Too vague. The AI has no idea what tone, length, style, or who it's for. You'll get something generic every time.

02

Add the Style

Tell the AI the **tone and feel** you want.

funny, happy, cute, cartoon, colorful

03

Add Details About the Person

Personalization makes everything better.

Include:

Their **name, age**, what they **love**

"Write a funny 8-line birthday poem for my friend Sam who loves soccer. Make it rhyme."

Specific, styled, and personal. The AI knows the **tone** (funny), the **format** (8 lines, rhyming), and the **person** (Sam, soccer fan).

See the difference? **More details = better result.**

Hands-On Activity — Make Your Birthday Surprise!

Time to make it! We'll use **ChatGPT** or **Google Gemini** to bring your idea to life.

01

Open the Tool

Go to **chat.openai.com** (ChatGPT) or **gemini.google.com** (Gemini)

02

Type Your Prompt

Use the prompt you wrote in Mini Activity 2

03

Look at What AI Made

Read it carefully — does it match what you imagined?

04

Make It Better

Change the prompt and try again — iterate!

05

Keep Going

Keep refining until you're happy with the result

✓ Remember: **YOU are the director. AI is your assistant.**



Try Your Prompt!

Some examples to try if you're stuck — use these as a starting point, then make them your own:



For a TEXT Prompt:

"Write a funny 8-line birthday poem for my friend [name] who loves [thing they love]. Make it rhyme and have a silly ending."



For an IMAGE Prompt:

"Create an image of a [dragon/unicorn/cat/dog] wearing a party hat, holding a birthday cake, in a magical forest. Style: cute cartoon, bright colors."



Try yours first. If it doesn't quite work — **change it and try again!** That's the whole point. Every change teaches you something new about how to talk to AI.

Look What You Made! 🎉

Let's talk about what happened when you tried your prompt:

Did the AI make what you imagined?

Was the result close to what you had in your head – or totally different?

AI gave you a starting point

What was close? What was off?

Which parts did AI nail? Which parts missed the mark?

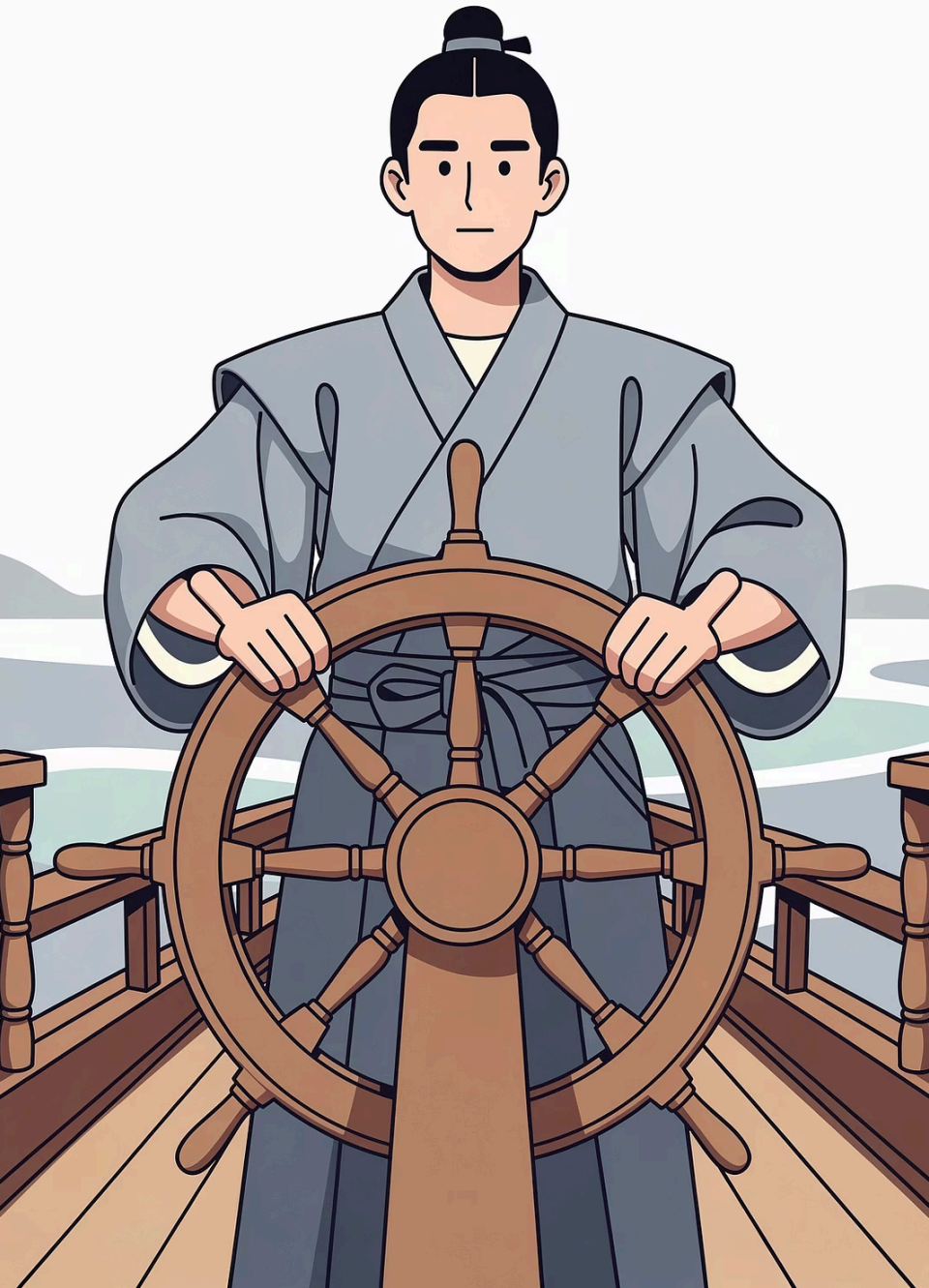
YOU shaped it

If you changed your prompt, what happened?

Did the result get better, worse, or just different?

The final result is YOURS

You just used Generative AI to create something brand new. That's powerful.



The Big Idea - You're the Decision Maker

Generative AI is amazing. But always remember who's really in charge:

AI gives you options → YOU pick what's good

AI generates many possibilities. Your judgment decides which one is right.

AI gives you drafts → YOU make it yours

AI creates a starting point. Your edits and direction make it meaningful.

AI gives you ideas → YOU bring the meaning

AI can't write a poem that means something to YOUR friend. Only you can.

AI helps. YOU create.

Your Homework – Make It Better

Take what you made this session and improve it. Here's exactly what to do:

01

Open the Tool

Open ChatGPT or Gemini at home

02

Try Again

Use your prompt again — but make it **even more specific**

03

Compare

Compare the new result to the one from class

04

Save the Best

Screenshot or copy the version you like most

Then discuss in next session:



What changed?

What did you change in your prompt?

What improved?

What was better in the new version?

What did you learn?

What did you discover about how to "talk" to AI?

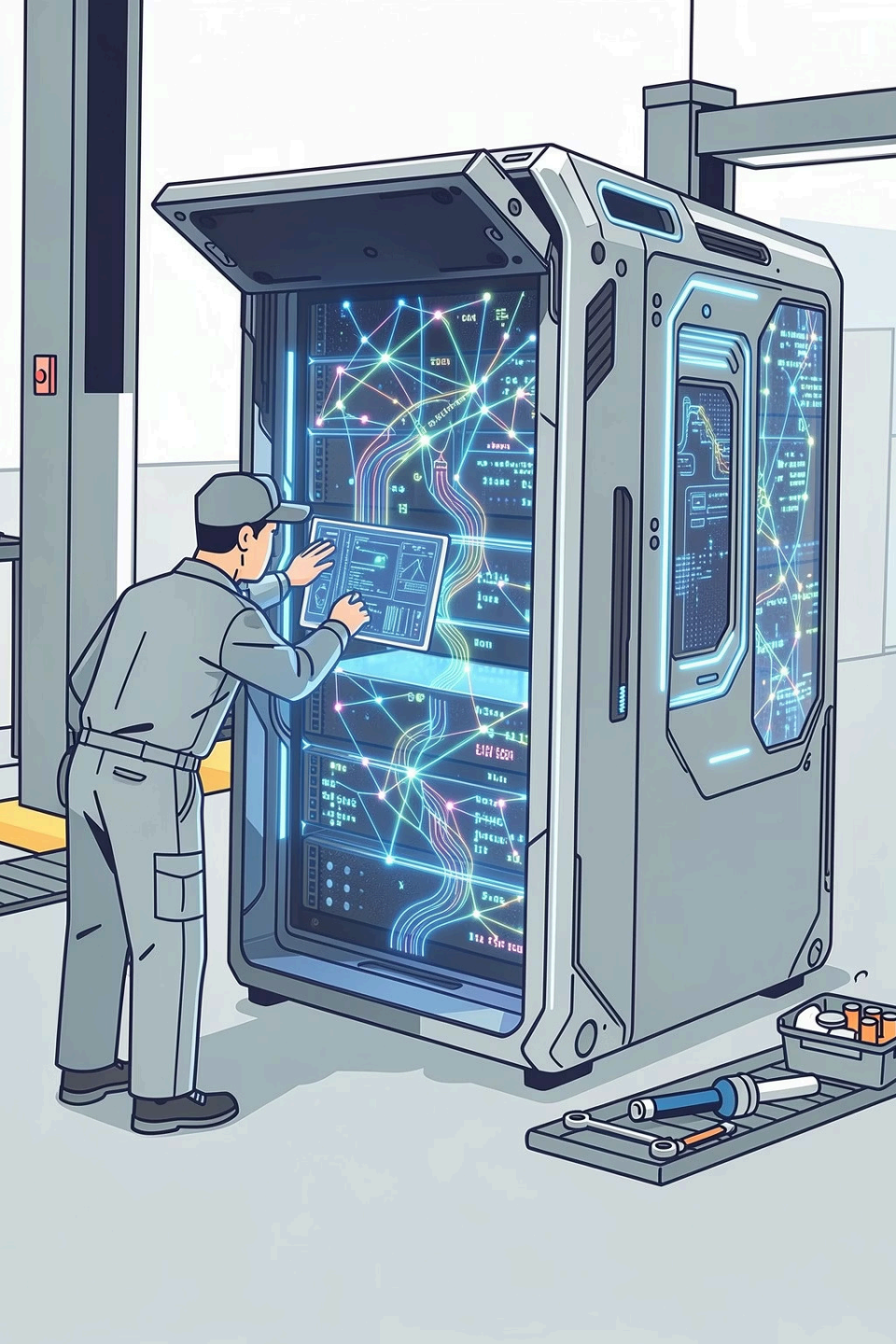
Challenge: Two Versions of the Same Thing ✨

If you want a stretch — try this extra challenge:

- 🎂 **Funny Birthday Poem**
Lighthearted, silly, full of jokes and puns — makes everyone laugh out loud.
- 🧡 **Heartfelt Birthday Poem**
Warm, sincere, emotionally resonant — moves people and creates lasting memories.
- 🐲 **Scary Dragon**
Fire-breathing, fearsome, shadowy — a creature of terror and ancient power.
- 🐉 **Cute Dragon**
Tiny, fluffy, wide-eyed — a lovable companion you'd want as a pet.

- ☑ See how much the prompt changes the result! This is the best way to really understand the power of prompting. One small word can completely transform what AI creates.





Next Time — Session 8: Large Language Models (LLMs)

Today you used ChatGPT and Gemini. But what's actually **INSIDE** them?
Next session, we open the hood:

What's a Language Model?

The engine behind every AI chatbot you've ever used

One Word at a Time

Why does AI predict one word at a time — and how does that work?

Sounds So Human

What makes ChatGPT sound so... human — even though it doesn't think?

Without Actually Thinking

How does it work without any real understanding?



Bring your improved prompt! We'll use it to explore how language models actually process your words.